



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Artificial Intelligence and Data Science

Academic Year: 2024- 2025 (ODD Semester)

Active Learning Best practices: Think Pair Share

Degree, Semester & Branch: III Sem. B.Tech. Artificial Intelligence and Data Science – 'A'

Course Code & Title: AD3391 – Database Design and Management

Name of the Faculty member: Mrs. G. Kavitha, AP/AI & DS

Theme of discussion: Think Pair Share

Topics Covered: Unit- III – Practical application of Normalization

Date and Time: 17.10.2024 & 09.45 A.M to 10.30 A.M

Active Learning Best practices: Think Pair Share

Unit / Topic: Unit III / Practical application of Normalization

Course Outcome: CO3 - Design database effectively by mapping ER model to Relational Model and applying Normalization.

Topic Learning Outcome: TLO31

Activity Chosen: Think-Pair-Share

Justification:

- Normalization is an important topic in Database Design and Management and also the question was asked in most of the university questions. Students think about a topic on their own and express their thoughts with their classmates, which helps them enhance their creative thinking, understanding the concepts and oral communication abilities.

Details of Implementation

- Active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs (Pair) or groups, and finally together with the entire class (Share).
- T (Think): Teacher asks a specific question about the topic. Students "think" about what they know or have learned, and come up with their own individual answer to the question. [Takes 1-3 Minutes].
- P (Pair): Teacher asks another question, related to the previous one that is suitable to deepen the students' understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task. [Takes 5-10 Minutes].

- S (Share): Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points. [Takes 10-15 minutes].

Time Allotted for the Activity: 30 Minutes

CO – PO / PSO Mapping

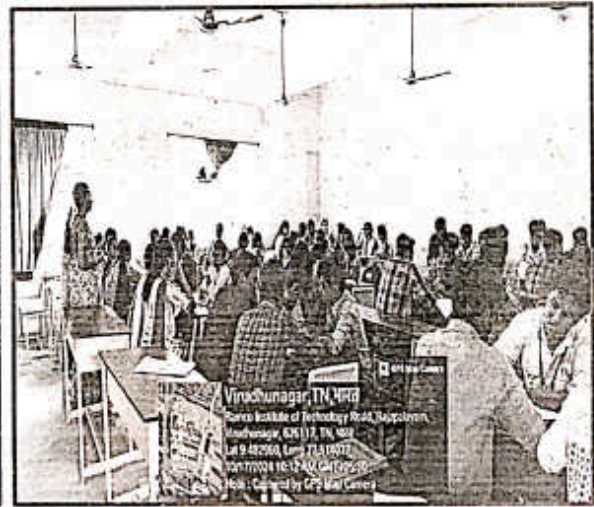
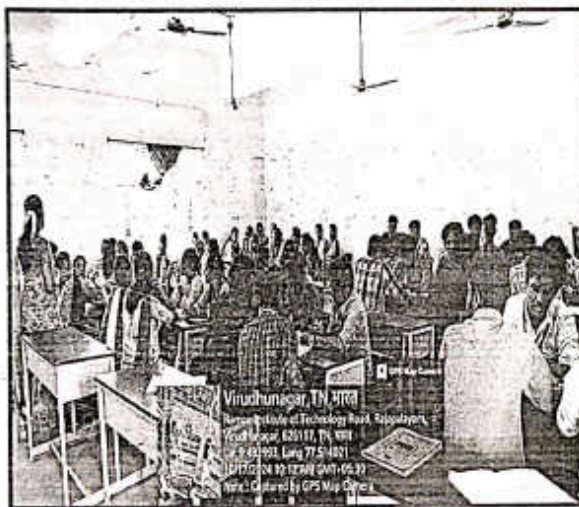
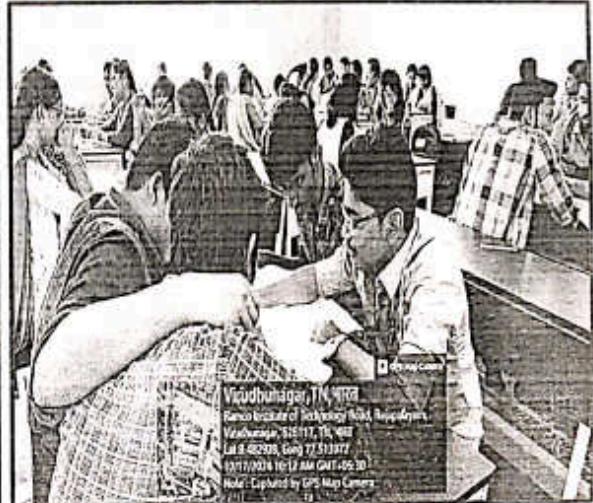
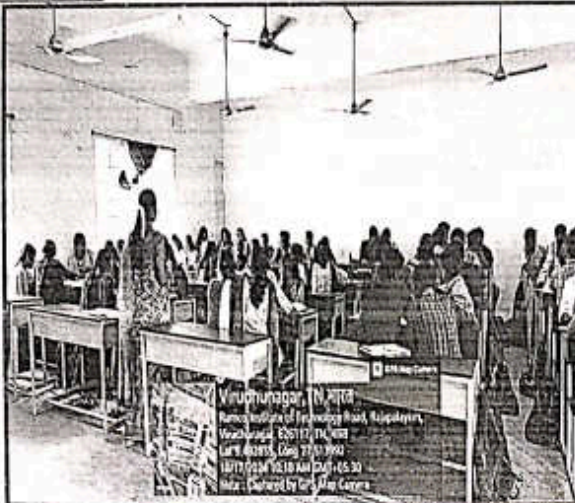
CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	2	2	2	2	2	2	2	2	3	2	3

PO/PSO Mapped

Innovative Practice	PO1 3	PO2 2	PO3 2	PO4 2	PO5 2	PO9 2
Justification for correlation	Students can be able to apply relational database design using normalization to solve engineering problems	Students can be able to identify requirements to create appropriate ER and EER models and map them to relational schemas using normalization.	Students can be able to design relational schemas to meet specified requirements.	Students can be able to investigate and prove functional dependencies using inference rules.	Students can be able to use database design tools (such as ERwin, Oracle SQL Developer Data Modeler, or Microsoft Visio).	Students can be able to analyze and resolve anomalies from multiple perspectives to ensure thoroughness and accuracy.

Innovative Practice	PO10 2	PO11 2	PO12 2	PSO1 3	PSO2 2	PSO3 3
Justification for correlation	Students can be able to describe the transformation process and the resulting relational schema to stakeholders.	Students can be able to manage the process of transforming ER/EER models to relational schemas by applying normalization within project timelines and budgets.	Students can be able to stay updated on best practices and tools for database design and mapping.	Students can be able to understand and apply fundamental principles of database design using normalization.	Students can be able to facilitate the migration and management of databases across hybrid and multi-cloud environments, ensuring consistent data integrity and performance using normalization.	Students can be able to create well-structured data from normalized databases which supports advanced analytics, predictive modeling, and data science initiatives, providing a competitive edge in data-driven decision-making.

Glimpses:



Reflective Critique

Feedback of practice from students and other stakeholders:

- Working as a pair, they can share their knowledge with them and come up with better understanding about the topic.
- Different types of reasoning/justification based questions are discussed in the class room, so students come to know how to answer/draw diagrams whenever new real time system is asked in the question.

Benefit of the practice:

- Students are actively engaged and students learn from each other.
- Makes class interactive and builds a friendly, yet academic atmosphere.
- Includes all the students in the teaching-learning process.

Abhishek
23/10/23

Signature of the Faculty Member

manu
23/10

HoD



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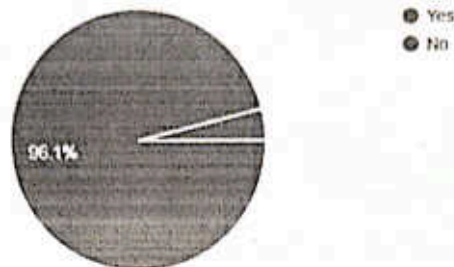
Feedback Questions:

1. Did the plan align with your expectations?
Yes No
2. How effectively was the innovative practice plan implemented?
Excellent Good Satisfactory
3. How well did you understand the objectives and purpose of the innovative practice plan?
1 2 3 4 5
4. Were the necessary resources and support provided for successful implementation?
Yes No
5. Did you notice any specific benefits or changes resulting from the plan?
Yes No
6. How engaged were you throughout the practice plan?
1 2 3 4 5
7. Did the plan provide opportunities for active participation and collaboration?
Yes No

Feedback Analysis:

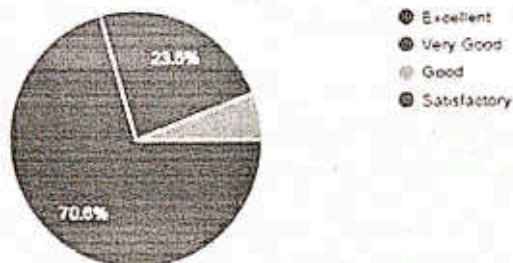
Did the plan align with your expectations?

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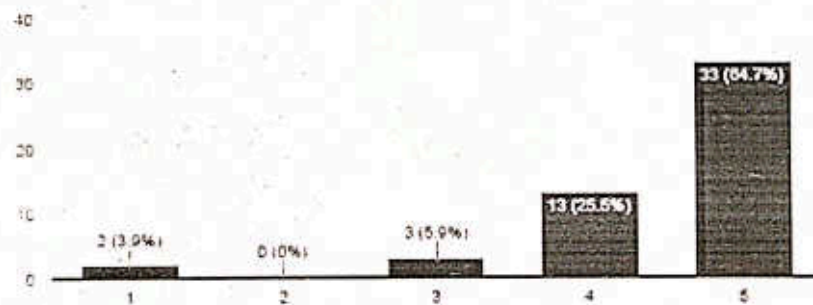
How effectively was the innovative practice plan implemented?

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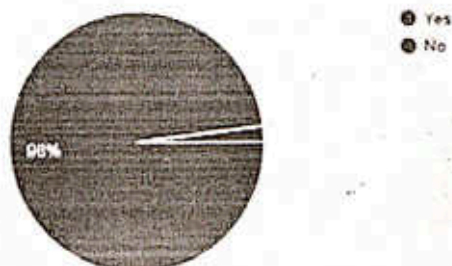
How well did you understand the objectives and purpose of the innovative practice plan?

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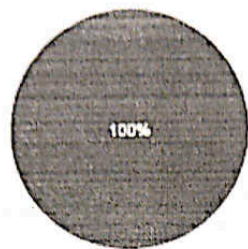
Were the necessary resources and support provided for successful implementation?

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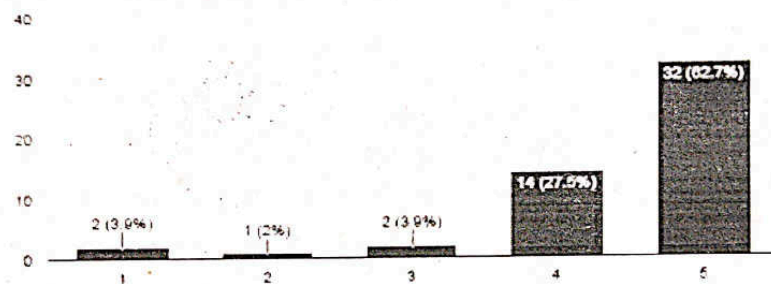
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● Yes
● No

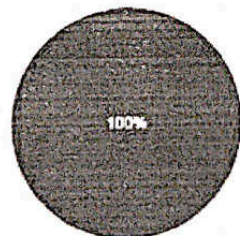
How engaged were you throughout the practice plan?

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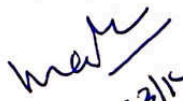
Did the plan provide opportunities for active participation and collaboration?

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● Yes
● No


Signature of the Faculty Member


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Name of the Faculty member: Mrs. G. Kavitha, AP/AI & DS

Theme of discussion: Think Pair Share

Topics Covered: Unit- III – Practical application of Normalization

Date and Time: 14.10.2024 & 02.40 P.M to 03.25 P.M

Active Learning Best practices: Think Pair Share

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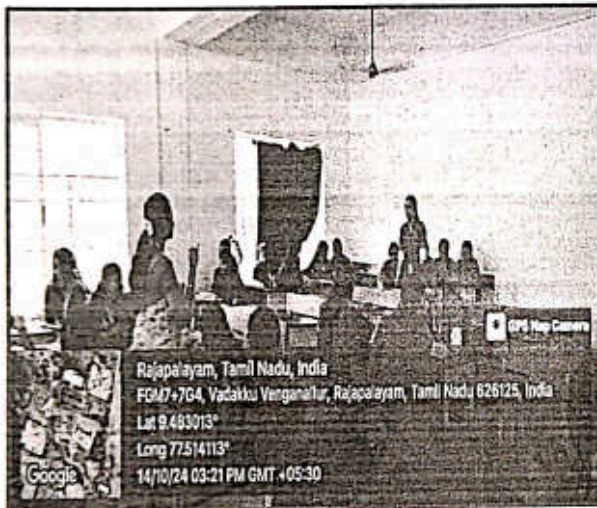
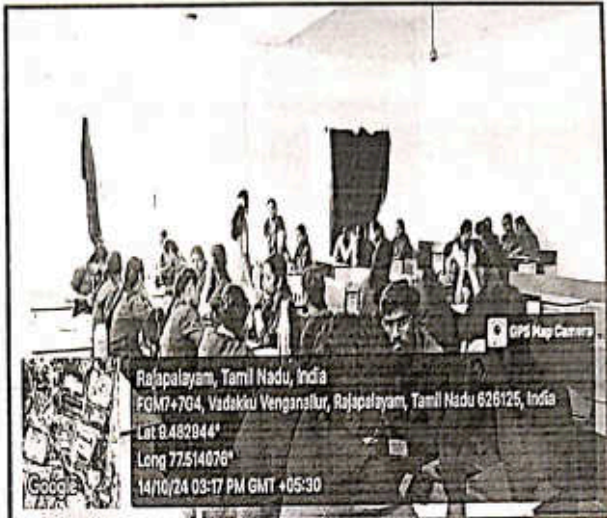
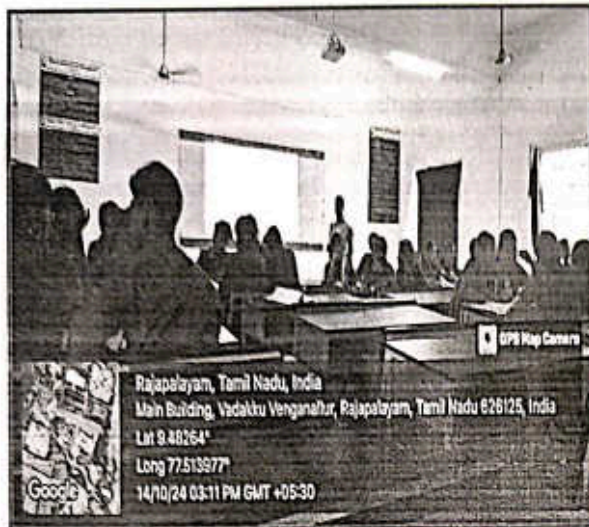
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[Signature]
Signature of the Faculty Member

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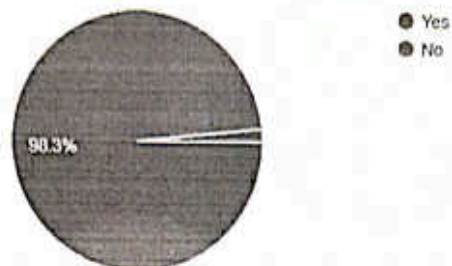
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Feedback Analysis:

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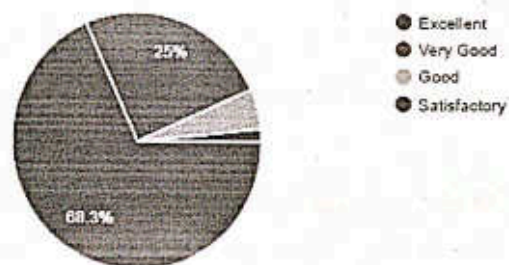
60 responses



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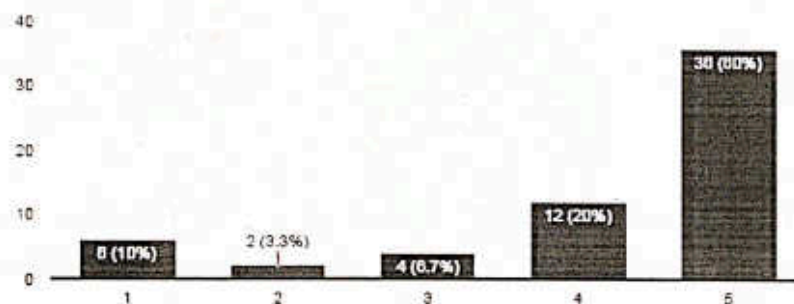
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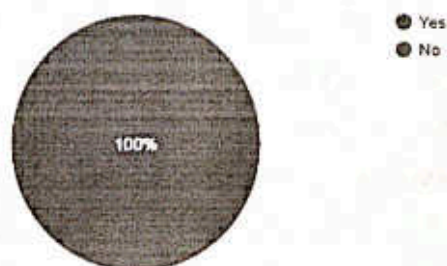
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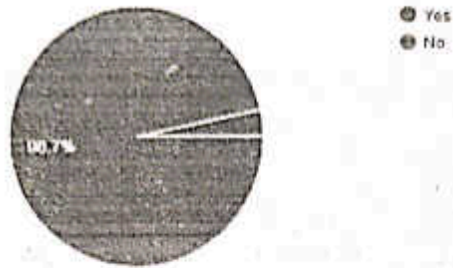
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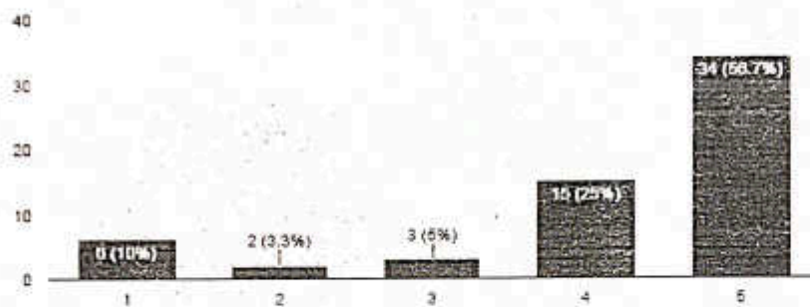
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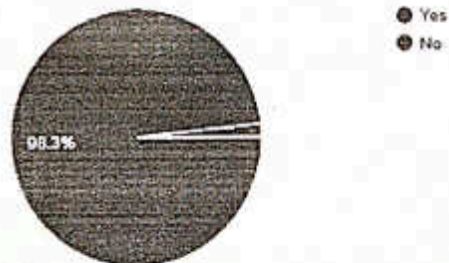
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